NIFTY Options IV Surface Reconstruction

Project Report

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1. Project Overview

The objective of this project was to reconstruct missing Implied Volatility (IV) values in a NIFTY 50 options chain dataset. Each row represents a 5-minute snapshot of the full option chain spanning the January 2026 expiry cycle (07 Jan - 27 Jan 2026). Approximately 20% of IV values were missing and needed to be imputed without using any future information (no lookahead bias).

Dataset at a glance:

- 975 timestamps (5-minute bars across 15 trading days)
- 28 option contracts: 14 Calls (CE) + 14 Puts (PE) across strikes 24400 - 26000
- Total observations: 27,300 & Missing: ~5,460 (~20%)
- Calls missing: ~19.7% & Puts missing: ~20.3% (near-symmetric, liquidity-driven)
- Expiry: 27 January 2026, 15:30 IST

Evaluation metric: Mean Squared Error (MSE) on hidden test cells.

2. Exploratory Data Analysis

2.1 Missing Data Profile

Missing values were not uniformly distributed. Three structural patterns emerged:

- Time-of-day clustering: highest missingness at 09:30 (market open auction) and 15:15 (position squaring before close)
- Strike / moneyness effect: deep Out-of-The-Money (OTM) wing strikes were consistently more illiquid
- Call-Put symmetry: when a Call was missing, the corresponding Put at the same strike was almost always missing too - confirming market-structure rather than data-quality causes

Missing rate by moneyness bucket:

Moneyness Bucket	Missing Rate
Deep OTM Put (< 0.92)	~21.4%
OTM Put (0.92 - 0.96)	~19.8%
NTM Put (0.96 - 1.00)	~18.2%
NTM Call (1.00 - 1.04)	~18.0%
OTM Call (1.04 - 1.08)	~19.6%

Moneyness Bucket	Missing Rate
Deep OTM Call (> 1.08)	~21.1%

2.2 Volatility Surface Structure

Visualising the observed IVs revealed the expected financial structure:

- Volatility smile / skew: IV forms a U-shape across strikes with ATM options at the minimum. Deep OTM puts carry elevated IV, reflecting equity crash-risk pricing.
- Time continuity: IV evolves gradually from day to day on normal trading days.
- Expiry Day regime: on 27 January 2026 (expiry), Gamma explodes near ATM. The smooth U-shape collapses into a sharp V-shape as time value approaches zero.
- ATM IV range across the cycle: approximately 0.10 - 0.18 on normal days, spiking to 0.30+ in deep OTM wings on expiry.

3. Modelling Approaches and Results

The following approaches were implemented keeping in mind to avoid lookahead bias.

3.1 Approach 1 - PCHIP Baseline + LightGBM Residual

Leaderboard MSE: 0.0027478

The standard residual learning strategy: fit a PCHIP cross-sectional interpolator as a baseline, then train a LightGBM model to correct the errors. This failed due to the Identity Function Trap - PCHIP passes exactly through observed data points, so residuals on observed cells are identically zero. The model learned to predict near-zero residuals everywhere and performed 10× worse than a plain interpolation baseline.

3.2 Approach 2 - Pure PCHIP + Temporal Forward Fill

Leaderboard MSE: 0.0112245

Dropped the ML corrector and used only PCHIP with unconstrained extrapolation. PCHIP oscillates when projecting beyond the observed strike range, generating unrealistic IV spikes at the wings. Additionally, a single regime was used for all days including Expiry Day.

3.3 Approach 3 - PCHIP Belly + Flat Wing Extrapolation

Leaderboard MSE: 0.0010454

Applied PCHIP only inside the observed range and replaced wing extrapolation with a flat carry of the edge IV. This is the standard market-maker convention for illiquid wing strikes. A 3× improvement over Approach 2, but still single-regime.

3.4 Approach 4 - Regime-Switching: PCHIP / Linear

Leaderboard MSE: 0.0004304

Introduced a two-regime pipeline: PCHIP for Jan 7-23 (U-shape), Linear interpolation for Jan 27 Expiry Day (V-shape). The regime switch delivered a 2.5× improvement, confirming that Expiry Day requires fundamentally different treatment.

3.5 Approach 5 - CV Framework + Regime Switching

Leaderboard MSE: 0.0010365

Built a 5-fold expanding-window cross-validation framework - the first approach with a reliable local validation signal. The CV loop revealed the exact magnitude of the Expiry Day problem. Note: the leaderboard score here was worse than Approach 4 because of a minor bug in the flat-wing fallback; the CV framework itself was the key contribution.

CV fold results (PCHIP regime):

Fold (Date)	Regime	Local CV MSE
Jan 20, 2026	Normal	0.000025
Jan 21, 2026	Normal	0.000016
Jan 22, 2026	Normal	0.000051
Jan 23, 2026	Normal	0.000060
Jan 27, 2026	Expiry Day	0.006417

The Expiry Day fold (Jan 27) is ~100× harder than normal days, driving disproportionate MSE.

3.6 Approach 6 - Natural Cubic Spline + CV

Leaderboard MSE: 0.0000431

Replaced PCHIP with a Natural Cubic Spline (bc_type='natural'). The natural boundary condition forces the second derivative to zero at both ends, causing the spline to extrapolate as a straight line into the wings - exactly matching market tail-risk pricing. More importantly, the natural spline does not flatten at the ATM minimum (PCHIP's weakness), preserving the curved U-shape belly.

3.7 Approach 7 - Weighted Polynomial + PCHIP Hybrid (Final Winner)

Leaderboard MSE: 0.0000337

The winning approach replaced the single global curve with a local, adaptive reconstructor. For each missing strike, a locally-weighted polynomial is fitted using only nearby observed strikes with Gaussian kernel weights. The bandwidth and polynomial degree are selected via leave-one-out cross-validation on the observed points in that cross-section.

Belly strikes (within observed range): 50% polynomial + 50% PCHIP blend

Wing strikes (outside observed range): 70/20/10 blend:

- 70% - polynomial anchored on same-side observed points
- 20% - polynomial with previously-filled wing values as additional anchors
- 10% - nearest-neighbour polynomial fallback

A temporal carry-forward dictionary (last_known_surface) provides warm-start fallbacks for timestamps where fewer than 2 observed strikes remain. All predictions are clipped to [1×10^{-6}, 6.0] to prevent physically impossible IVs.

4. Full Results Summary

Approach	Method	LB MSE
1	PCHIP Baseline + LightGBM Residual	0.0027478
2	Pure PCHIP + Temporal Forward Fill	0.0112245
3	PCHIP Belly + Flat Wing Extrapolation	0.0010454
4	Regime-Switching: PCHIP / Linear	0.0004304
5	Regime PCHIP / Linear + CV Framework	0.0010365
6	Natural Cubic Spline + CV	0.0000431
7	Weighted Poly + PCHIP Hybrid	0.0000337

Local CV MSE (20%-masking, random held-out cells): 0.00008021 for Approach 7, confirming the improvement generalises beyond the leaderboard split.

5. Key Lessons

- Mathematics beats Machine Learning for smooth manifolds
- Regime awareness is mandatory
- Local adaptation beats global fitting
- Wing extrapolation is the decisive sub-problem